Code: DMB-01 **Paper ID: 13441**

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)

(TRIMESTER –III) END TERM EXAMINATION, MAY -2025

ANALYTICS FOR MANAGERS USING PYTHON

[Time: 2 Hours] *[Maximum Marks: 60]*

Following are the course outcomes of the subject: - ANALYTICS FOR MANAGERS USING PYTHON

CO	Course Outcome (CO)
CO1	Importing data from diverse sources, explore datasets and ensuring effective data manipulation.
CO2	Expertise in descriptive statistics, visualization methods and correlation analysis, enabling them to draw meaningful insights from data using Matplotlib and Seaborn.
CO3	Develop skills in creating compelling visualizations, facilitating effective data storytelling and communication.

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)

Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	**4**	CO - 1	BL1
(B)	**4**	CO - 1	BL1
(C)	**4**	CO - 1	BL2
(D)	**4**	CO - 1	BL2
(E)	**4**	CO - 1	BL1

SECTION-B

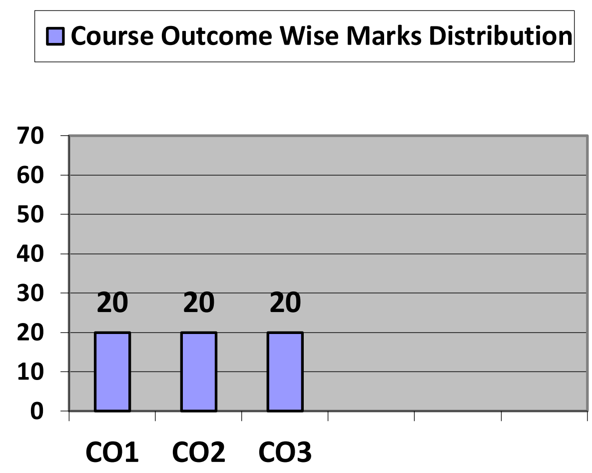
Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)

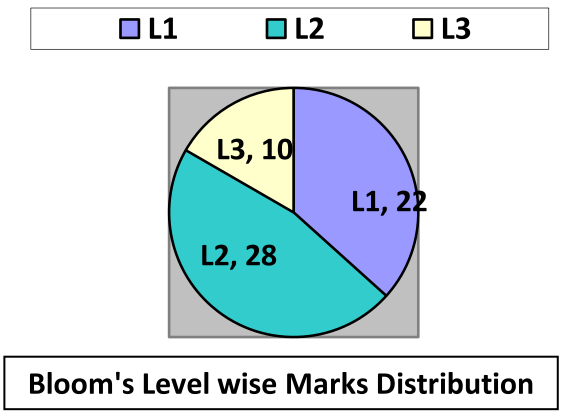
Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	**10**	CO – 2	BL3
2 (B)	**10**	CO – 2	BL3
3 (A)	**10**	CO – 2	BL2
3 (B)	**10**	CO – 2	BL2

SECTION-C

Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	**10**	CO – 3	BL1
4 *(B)*	**10**	CO - 3	BL2





Moderated By **IQAC** **Dean - Examinations**

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[Time: 2 Hours] *[Maximum Marks: 60]*

INSTRUCTIONS:

(i) There are THREE sections in this question paper.
(ii) Attempt questions from each section as per the directions.
(iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: *(5×4=20)*

(a) Why is Python preferred for Data Analytics?
(b) What are the main Python libraries used in Data Analytics?
(c) Describe the methods by which you write in .csv files using python and discuss their properties.
(d) Explain the concept of Data Frames in Pandas
(e) What is the role of NumPy in Data Analytics?

SECTION-B

(Long Answer Type Questions)

Q2. Attempt all questions of the following *(2X10=20)*

(a) for the following dataset, apply isna method for all possible cases.

	Student	Class	Year	Grade
0	Emma	Freshman Seminar	Freshman	86.0
1	Olivia	Freshman Seminar	Freshman	86.0
2	Noah	Freshman Seminar	Freshman	86.0
3	Sophia	Freshman Seminar	Freshman	87.0
4	Liam	Freshman Seminar	Freshman	90.0
...	...	...	...	...
81	NaN	NaN	NaN	NaN
82	Bennett	NaN	NaN	NaN
83	NaN	EDA	Junior	84.0
84	Gavin	EDA	Senior	NaN
85	Calvin	NaN	NaN	100.0

OR

(b). Case Study: Analysing Sales Data for a Retail Company

Y Background:

You are a data analyst at a mid-sized retail company, **RetailX**, which operates across multiple cities. Your manager has provided you with a CSV file named retail_sales.csv. The file contains sales data for the year 2024 and includes the following columns:

- InvoiceID – Unique identifier for each transaction
- Date – Date of the transaction
- CustomerID – Unique customer ID
- City – City where the purchase was made
- ProductCategory – Category of the product sold

- Quantity – Number of items sold
- UnitPrice – Price per unit of the product
- PaymentMethod – Method used to make the payment
- DiscountApplied – Discount applied on the transaction (in %)

1. Load the dataset into a Pandas DataFrame. Display the first 5 rows.
   Check for missing values in the dataset. How many missing values are in each column?
   Convert the `Date` column to `datetime` type
2. How would you track sales trends over the year? What visualizations would be most effective to display seasonality?
3. How would you identify cities where certain product categories perform exceptionally well or poorly?

Q3. (a) For the following csv dataset, write the Python statements for reading Book_Customers file & illustrate the use of separator & header

	Customer ID	Customer Name	Age	Has School Aged Children	Has_Pets
1	101	Alexander	36	Yes	No
2	102	Mason	16	No	No
3	103	Ethan	55	No	No
4	104	Daniel	70	No	No
5	105	Michael	18	No	Yes

Your manager wants to see a visual representation of **customers' ages** in a **bar chart**, using their **names on the x- axis**. You're expected to:

1. Describe how you would prepare the data for the bar chart.
2. Write or explain Python code that generates the bar chart.
3. Mention one potential issue when using customer names on the x-axis and how to solve it.

OR

*(b).*1. Explore the data frame using below file spotify_songs.csv& find out its basic information, number of rows & columns, no. of values in each column, display non-null values, top & bottom 5 rows

2. Explain theoretically how missing values in the Popularity column could affect analysis

	name	artists	energy	key	mode	valence	tempo
1	God's Plan	Drake	0.449	7	1	0.357	77.169
2	SAD!	XXXTENTACION	0.613	8	1	0.473	75.023
3	ockstar (feat. 21 Savage)	Post Malone	0.535	5	0	0.14	159.847
4	;ycho (feat. Ty Dolla $ign)	Post Malone	0.559	8	1	0.439	140.124
5	In My Feelings	Drake	0.626	1	1	0.35	91.03
6	Better Now	Post Malone	0.563	10	1	0.374	145.028
7	I Like It	Cardi B	0.726	5	0	0.65	136.048
8	One Kiss (with Dua Lipa)	Calvin Harris	0.862	9	0	0.592	123.994
9	IDGAF	Dua Lipa	0.544	7	1	0.51	97.028
10	FRIENDS	Marshmello	0.88	9	0	0.534	95.079
11	Havana	Camila Cabello	0.523	2	1	0.394	104.988
12	Lucid Dreams	Juice WRLD	0.566	6	0	0.218	83.903
13	Nice For What	Drake	0.909	8	1	0.757	93.394
14	ls Like You (feat. Cardi B)	Maroon 5	0.541	0	1	0.448	124.959
15	The Middle	Zedd	0.657	7	1	0.437	107.01
16	All The Stars (with SZA)	Kendrick Lamar	0.633	8	1	0.552	96.924
17	no tears left to cry	Ariana Grande	0.713	9	0	0.354	121.993

SECTION-C

Q4. Attempt all parts of the following: ***(2×10=20)***

Customer Behavior & Sales Analysis Using Plotly

Background:

You are a data analyst at **ShopEase**, a rapidly growing online retailer. The marketing and product teams want an **interactive dashboard** that gives insight into customer behavior, spending trends, and product performance.

You've received a cleaned dataset shop_data.csv with the following fields:

The dataset contains over **10,000 rows of transaction-level data** from **January to December 2024**.

Column Name	**Description**
CustomerID	Unique identifier for each customer
CustomerName	Full name of the customer
Age	Customer's age
Gender	Gender of the customer
City	City where the customer lives
ProductCategory	Type of product purchased
AmountSpent	Amount spent in a single transaction
PurchaseDate	Date of the transaction (format: YYYY-MM-DD)

1. 1. How would you prepare this dataset for visualization with Plotly?
2. 2. How would you show the top 10 customers by total spending using a Plotly bar chart?
3. How would you visualize the relationship between age and amount spent, colored by gender

(b) You are tasked with building an interactive dashboard using Dash.

1. 1. Explain how callbacks work in Dash and how you would use them in this case.
2. 2. Discuss the benefits of using Dash compared to static plots in reporting.
3. 3. Outline performance or deployment considerations for large datasets in Dash.